II श्री गणेश प्रसन्न II

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · Innate and adaptive immunity

All immunotherapy acts on these arms.

- P2 Describe use and principles of immunology and Immunotherapy in Urological Oncology — *innate/adaptive arms, TLR/PAMP, two signals, Th1 vs Th2, checkpoints, adaptive immune resistance, three Es*

Not in the paper answer:

- Serum IL-2, CXCL8, CCR7/CCL21 homing, class II peptide length, CPS <10 (all below).
- Bacille Calmette-Guérin (BCG) cell-wall peptidoglycans are pathogen-associated molecular patterns (PAMPs), binding **Toll-like receptor 2 (TLR2)** on bladder innate cells.
- The sentinel urothelium makes type I interferon (IFN), raising class I major histocompatibility complex (MHC), and CXCL8 (interleukin-8, IL-8), attracting neutrophils.
- The **regulatory T cell (Treg) is a CD4 subset, marker FoxP3**.
- T-helper 1 (Th1) cytokines (IL-2, tumour necrosis factor-α [TNF-α], IFN-γ) mean **BCG success**; Th2 (IL-4, IL-5, IL-10, IL-13) means **BCG failure**. Raised serum IL-2 after BCG is favourable.

ALGORITHM

1. A danger signal (TNF-α, TLR ligand) makes the dendritic cell (DC) stop antigen uptake and raise MHC and B7-1/B7-2.
2. It travels via afferent lymphatics (CCL21 sensed by CCR7) and activates CD4, CD8 and B cells in the node.

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; raised by inflammation and IFN	Antigen-presenting cells (APCs)
Peptide	~9 amino acids	13–25 amino acids

2 · Immune editing and escape

Clinical cancer has escaped; treatment reverses this.

- Escape means losing antigen or MHC and gaining Tregs; prior equilibrium may last a lifetime.
- Adaptive immune resistance: CD8 IFN-γ induces programmed death ligand-1 (PD-L1) on tumour and myeloid cells, which binds programmed death-1 (PD-1) on the same CD8 cells and switches them off.

KEY POINT

PD-L1 is largely induced by the immune attack itself — a marker of an ongoing but suppressed response, not a fixed tumour trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating

T-cell molecule	Ligand	Effect
Cytotoxic T-lymphocyte antigen-4 (CTLA-4)	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
Lymphocyte activation gene-3 (LAG-3)	MHC class II	Inhibitory

3 · Inflammation and GU tumours

Each genitourinary (GU) cancer's immune setting predicts which drugs work.

	Bladder	Kidney	Prostate
Inflammation causes cancer?	Strongly: *S. haematobium* (squamous), chronic urinary infection, catheters, cyclophosphamide	Weakly (smoking, obesity)	Probably: peripheral-zone inflammation; PIA beside HGPIN
Effect of CD8 infiltration	Better outcome (muscle-invasive bladder cancer, MIBC)	**Worse — renal paradox**	CD8 cells may be suppressive
Dominant restraint	PD-L1, rising with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

- Chronic Th2/Treg/M2-macrophage inflammation promotes cancer; acute Th1/CD8 response eliminates it.

4 · BCG in bladder cancer

BCG is the most successful urological immunotherapy.

> **ALGORITHM**
>
> 1. Instil 1–5 × 10^8 bacilli; some adhere via **fibronectin attachment protein**.
> 2. TLR2/TLR4 on urothelium release IL-8 and TNF-α; a neutrophil wave (hours) activates DCs.
> 3. A nodal CD4 Th1 response (weeks) leaves memory.

- Memory is implied as ~**50%** of non-muscle-invasive bladder cancer (NMIBC) remains in durable remission.
- **Nadofaragene firadenovec**: IFN-α2b protein is not durable, so a non-replicating adenovirus carries its gene, plus **Syn3** to overcome the antiadherence barrier.
- Given 3-monthly up to 12 months, it achieves carcinoma in situ (CIS) complete response (CR) of 53.4%. The Food and Drug Administration (FDA) approved it in 2022 as the **first gene therapy** for BCG-unresponsive NMIBC with CIS.

5 · Cancer vaccines

Only one vaccine has succeeded in phase III.

- P4 Rationale of using cancer vaccines in Urology — *antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC*

Not in the paper answer:

- Nothing important.

- Sipuleucel-T improves overall survival (OS) by **+4.1 months** but has **no effect on time to progression**. It was the first FDA-approved solid-tumour vaccine.

6 · Checkpoint blockade

Checkpoint drugs work in bladder and kidney, rarely prostate.

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052) is only for the cisplatin-ineligible with combined positive score (CPS) **≥10**, or the platinum-ineligible. CPS <10 did worse than carboplatin.
- Atezolizumab and durvalumab were withdrawn (failed confirmatory trials); adjuvant atezolizumab (IMvigor010) failed.
- **Enfortumab vedotin (EV)** is an antibody–drug conjugate (anti-nectin-4 + monomethyl auristatin E). With pembrolizumab, objective response rate (ORR) was 68% in the cisplatin-ineligible (EV-103).
- †EV-302 roughly doubled progression-free survival (PFS) and OS versus platinum, so it is first line regardless of cisplatin eligibility.

Kidney

- CTLA-4 is on Tregs and PD-1 on CD8 TILs, so dual blockade is synergistic.
- Vascular endothelial growth factor (VEGF) blocks DC maturation and T-cell entry. Tyrosine kinase inhibitors (TKIs) normalise vessels: the rationale for TKI plus immunotherapy.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- In metastatic castration-resistant prostate cancer (mCRPC), pembrolizumab gives PSA response <10%, as the tumour is rarely PD-L1-positive.
- Ipilimumab (two phase III trials) and atezolizumab plus enzalutamide (IMbassador250; enzalutamide raises PD-L1) failed to improve OS.
- Exception: **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient; 1.2–12%) tumours have many neoantigens and get pembrolizumab second line or beyond.
- Its 2017 tumour-agnostic approval was the **first based on biomarker, not site**.
- †Immune-related adverse events (colitis, hepatitis, pneumonitis, endocrinopathies): for grade ≥2, hold the drug and give corticosteroids; endocrinopathies need lifelong replacement.

KEY POINT

Every successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, immunogenic cell death, VEGF inhibition) or removes a brake (PD-1, CTLA-4). Failures cluster where the brake is not limiting — PD-L1-negative, Treg-dominated prostate cancer.

7 · Landmark trials

These define current standards.

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only anti-PD-1/PD-L1 approved on phase III second-line data
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	Disease-free survival 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, metastatic renal cell carcinoma (RCC) after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab 1 year vs placebo, high-risk clear-cell RCC	Recurrence/death reduced 32%	Adjuvant standard